

Churn Prediction Modeling in Telecom Sector Using Python

Project Proposal Submitted

To

Dr. D. Y. Patil Vidyapeeth, Centre For Online Learning, Pune



BY

Aryan Raina

241114835

240502007175

Specialization

Artificial Intelligence & Machine Learning

Name of the internal Supervisor

Professor Dr. Mahantesh M. Kodabagi

Ph.D.

Dr. D. Y. Patil Vidyapeeth, Centre For Online Learning, Pune

2024-2026

Sr. No	Index	Page no
1.	Introduction	3
2.	Objectives of the Study	3
3.	Scope of the Project	4
4.	Work Plan and Methodology	5
5.	References	8

1. Introduction

The telecom industry is very competitive these days. So customer retention is a challenge for service providers. Telecom companies spend a lot of money to get customers.. If they lose existing customers it can affect their revenue and long-term business growth. Customer churn is when customers stop using a companys services and switch to another service provider. So predicting customer churn is essential for telecom organizations to keep customers loyal and make money.

With data analytics and machine learning companies can now look at customer behavior patterns. Find out which customers are likely to leave. Machine learning models can help businesses take action before its too late by understanding what causes customer churn. Customer information like tenure, monthly charges, internet services, payment methods and contract types can give us insights for prediction.

This project is about building a customer churn prediction model using machine learning techniques in Python. The project includes data preprocessing, exploratory data analysis, hypothesis testing, feature engineering, model building and performance optimization. We compare machine learning algorithms to find the most effective model for churn prediction. The project also uses techniques like SMOTE to handle data and hyperparameter tuning to improve model performance. The findings of this study can help telecom companies find high-risk customers and improve customer retention strategies.

2. Objectives of the Study

1. We want to study customer churn behavior in the telecom sector using customer data.
2. We need to clean and preprocess the data and do exploratory data analysis to understand churn patterns.
3. We have to apply hypothesis testing to find out what factors affect customer churn.
4. We will Compare different machine learning models for churn prediction.

5. We want to improve model performance using feature engineering, SMOTE and hyperparameter tuning.
6. We need to find out what variables are most important for customer churn prediction.

3. Scope of the Project

This project is about customer churn prediction in the telecom sector using machine learning. We use a telecom customer dataset with customer details, account information, services subscribed and billing-related information.

We focus on analyzing customer behavior and predicting whether a customer will churn or not. We use machine learning models and evaluate them based on performance metrics like Accuracy, ROC-AUC Score, Precision, Recall and F1-Score.

The project is developed using Python and data science libraries like Pandas, NumPy, Scikit-learn, XGBoost, LightGBM and CatBoost.

This study can help telecom companies improve customer retention find high-risk customers reduce revenue loss and make data-driven decisions.

However the project has some limitations:

- We only use one dataset.
- We do not deploy the model in time.
- We do not consider market conditions and customer psychology.
- The prediction only depends on structured data.

4. Work Plan and Methodology

The project follows a machine learning workflow for customer churn prediction.

Phase 1: Data Collection

We use the Telecom Customer Churn dataset. The dataset has customer-related information like:

- Gender
- Tenure
- Monthly Charges
- Total Charges
- Internet Services
- Contract Type
- Payment Method
- Churn Status

We import the dataset into Python for analysis and model development.

Phase 2: Data Cleaning and Preprocessing

We preprocess the data to improve its quality and prepare it for machine learning models.

We do the following steps:

- Handle missing values
- Convert data types
- Remove columns
- Encode categorical variables
- Scale features using StandardScaler

We use encoding techniques like binary encoding and one-hot encoding based on feature types.

Phase 3: Exploratory Data Analysis (EDA)

We do exploratory data analysis to understand customer behavior and find patterns that affect churn.

We do the following analyses:

- Churn distribution analysis
- Tenure analysis
- Monthly charges analysis
- Contract type analysis
- Internet service analysis
- Correlation analysis

We use graphs and visualizations to understand the data.

Phase 4: Hypothesis Testing

We use hypothesis testing to find out if certain customer attributes significantly affect churn behavior.

We test relationships between:

- Contract type and churn
- Internet services and churn
- Payment methods and churn
- Monthly charges and churn

The analysis helps us find variables that influence customer retention.

Phase 5: Feature Engineering

We do feature engineering to improve the capability of machine learning models.

We create features like:

- Average Monthly Spend
- Long-Term Customer Indicator
- High Monthly Charges Indicator

These features help the models capture additional customer behavior patterns.

Phase 6: Handling Imbalanced Data

The customer churn dataset is imbalanced because there are more non-churn customers than churn customers.

We use SMOTE to balance the dataset by generating samples for the minority class.

This improves the models ability to predict churn customers effectively.

Phase 7: Machine Learning Model Building

We. Compare different machine learning algorithms for churn prediction.

We use the following models:

- Logistic Regression
- Decision Tree
- Random Forest
- K-Nearest Neighbors (KNN)
- Support Vector Machine (SVM)
- Bayes
- XGBoost
- LightGBM
- CatBoost

Each model is trained and evaluated using different performance metrics.

Phase 8: Model Evaluation and Optimization

We evaluate the machine learning models using:

- Accuracy Score
- ROC-AUC Score
- Precision
- Recall
- F1-Score
- Confusion Matrix

We do hyperparameter tuning using GridSearchCV to improve model performance.

We also do threshold tuning to improve recall and F1-score for churn prediction.

Phase 9: Final Model Selection

We select the performing model as the final churn prediction model based on model comparison and evaluation metrics.

We also do feature importance analysis to find out what variables have the impact on customer churn.

5. References

1. Han, J., Kamber, M., & Pei, J. *Data Mining: Concepts and Techniques*. Morgan Kaufmann Publishers.
2. Géron, A. *Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow*. O'Reilly Media.
3. Pedregosa, F. et al. "Scikit-learn: Machine Learning in Python." *Journal of Machine Learning Research*, vol. 12, 2011.
4. Chawla, N. V. et al. "SMOTE: Synthetic Minority Over-sampling Technique." *Journal of Artificial Intelligence Research*, vol. 16, 2002.
5. Kaggle Telecom Customer Churn Dataset.
6. Official Documentation of:
 - Python
 - Scikit-learn
 - XGBoost

- LightGBM

- CatBoost

7. Brownlee, J. *Machine Learning Mastery with Python*. Machine Learning Mastery Publications.